

1 Baseline Covariate Adjustment in Randomized Clinical Trials:
2 Objective Criteria for When and How to Adjust

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38 1 Abstract

39 **Background.** Covariate adjustment in randomized clinical trials remains methodologically
40 contested in applied practice despite a mature theoretical literature, mature regulatory guid-
41 ance from FDA, EMA, and ICH E9, and consistent simulation evidence. The disconnect
42 between literature and practice is partly attributable to the absence of objective criteria that
43 an investigator can apply without re-deriving the relevant theory; this paper develops such
44 criteria.

45 **Methods.** We synthesize the statistical literature on covariate adjustment in RCTs, the rele-
46 vant regulatory guidance, and illustrative simulation evidence on adjusted- versus-unadjusted
47 analyses across varying covariate-outcome correlations and sample sizes. Performance mea-
48 sures include bias, coverage, and efficiency gain (relative variance reduction), reported with
49 Monte Carlo standard errors following the ADEMP framework of¹. The synthesis distills into
50 objective criteria along three axes: when to adjust, which covariates to adjust for, and how
51 to specify the adjustment.

52 **Results.** The criteria converge: adjustment should always be applied when prognostic co-
53 variates are pre-specified; stratification variables used in the randomization must appear as
54 covariates in the analysis; the baseline measurement of the outcome variable should always
55 be included when available. Selection of additional covariates should rest on prior evidence of
56 prognostic value, with the number of covariates kept small relative to sample size. Decisions to
57 adjust must not depend on observed baseline balance, a principle established by². Across the
58 simulation grid, ANCOVA dominates the unadjusted analysis on efficiency without compro-
59 mising coverage, while the test-then-adjust strategy shows systematic over-coverage relative
60 to nominal 95%. A second scenario with a binary outcome shows that marginal standard-
61 ization (g-computation) remains approximately unbiased for the true marginal risk difference
62 even when its working model omits a true nonlinear covariate term, at a modest efficiency
63 cost relative to the correctly specified working model.

64 **Conclusions.** The objective criteria provide an applied investigator with a procedural route
65 from trial design to analysis-model specification that aligns with the existing theoretical
66 and regulatory consensus. The criteria are not new findings but are a consolidation of dis-
67 persed guidance into practitioner-usable form, with explicit citation of the underlying primary
68 sources; this paper is a tutorial/review, not a report of novel methodology.

2 Introduction

Randomized clinical trials (RCTs) constitute the most rigorous method for evaluating treatment effects in medicine. A fundamental analytical decision in every RCT is whether and how to adjust for baseline covariates in the primary analysis. Despite decades of statistical research and increasingly clear regulatory guidance, confusion persists. Investigators routinely test for baseline balance between treatment groups (and adjust only if imbalances are detected), omit covariates from the primary analysis to preserve ‘simplicity,’ or select covariates post hoc based on observed associations with the outcome. Each of these practices is suboptimal or outright harmful. We review the theoretical, empirical, and regulatory evidence bearing on covariate adjustment and synthesize it into a set of objective criteria for when and how to adjust.

2.1 The Problem of Baseline Imbalance

Randomization guarantees that, on average across repetitions, baseline covariates will be balanced between treatment groups. It does not guarantee balance in any individual trial.³ demonstrated that even non-significant imbalances in prognostic factors can strongly influence the observed treatment effect, rendering conclusions misleading. The practice of testing for baseline balance and adjusting only when a difference is ‘significant’ was criticized sharply by², who argued it is ‘philosophically unsound, of no practical value and potentially misleading.’ Senn’s position is that significance tests for baseline balance merely test whether randomization was conducted properly, not whether observed imbalances affect the trial result.

Surveys of published trials confirm that these problematic practices remain widespread.⁴ examined 50 trial reports in four major journals, finding that most trials emphasized simple unadjusted results, that covariate adjustment usually made negligible difference, and that subgroup analyses were routinely overinterpreted. A companion paper by the same group⁵ highlighted the underuse of appropriate interaction tests and the inconsistency with which covariates were selected and adjusted for across studies.⁶ showed that among trials using stratified randomization, only 14 of 41 eligible studies adjusted for the stratification variables, leading to overly conservative inference with inflated confidence intervals and reduced power.

2.2 Theoretical Foundations

2.2.1 Analysis of Covariance in the Linear Setting

The classical statistical argument for covariate adjustment rests on the analysis of covariance (ANCOVA). When the outcome Y depends on a baseline covariate X and treatment Z , the ANCOVA model

$$Y_i = \beta_0 + \beta_1 Z_i + \beta_2 X_i + \epsilon_i$$

yields a more precise estimate of β_1 than the unadjusted comparison of means, provided X is prognostic for Y . The variance reduction is proportional to $R_{Y|X}^2$, the proportion of outcome variance explained by the covariate. This result holds regardless of whether randomization produced balanced or imbalanced groups^{7,8} showed that ANCOVA is superior to the analysis

107 of change scores (the ‘simple analysis of change scores,’ or SACS) and that ANCOVA does not
108 require baseline balance, even in expectation, to provide unbiased treatment effect estimates.
109 This resolves Lord’s paradox in the clinical trial setting⁸.

110 ⁹ provided a concise tutorial confirming that ANCOVA is the preferred analysis whenever a
111 baseline measurement of the outcome exists, yielding estimates with smaller standard errors
112 and greater statistical power than either the comparison of follow-up values alone or the
113 analysis of change scores.

114 ¹⁰ formally studied the relative efficiency of the change-score estimator, the follow-up-only
115 estimator, and several ANCOVA variants in the context of a pretest-posttest trial. They
116 showed that ANCOVA with a treatment-by-covariate interaction term (ANCOVA II) and
117 the GEE approach are most efficient, while even the simpler ANCOVA without interaction
118 dominates both the change-score and follow-up-only estimators for all values of the baseline-
119 outcome correlation.

120 **2.2.2 The Freedman-Lin Debate**

121 ¹¹ challenged the use of OLS regression adjustment in randomized experiments under Ney-
122 man’s randomization inference framework. He showed that adjustment can worsen asymptotic
123 precision, that standard errors from the OLS model are not generally valid, and that small-
124 sample bias exists. These results generated considerable concern.

125 ¹² reexamined Freedman’s critique and showed that the essential problem was the omission
126 of treatment-by-covariate interactions. When a full set of treatment-covariate interactions
127 is included in the regression, OLS adjustment cannot hurt asymptotic precision, and the
128 Huber-White sandwich variance estimator provides valid standard errors. In balanced two-
129 group designs, even the simpler model without interactions is asymptotically valid because,
130 as Lin put it, ‘two wrongs make a right’ – the overadjustment for one group cancels the
131 underadjustment for the other.

132 **2.2.3 Semiparametric Efficiency Theory**

133 The most rigorous theoretical framework for covariate adjustment in RCTs comes from semi-
134 parametric efficiency theory.¹³ and¹⁴ developed a class of semiparametric estimators for the
135 average treatment effect that are guaranteed to be at least as efficient as the unadjusted
136 estimator and typically much more so. The key insight is that in a randomized trial, any
137 function of the baseline covariates can be incorporated into the estimator to reduce variance
138 without introducing bias, provided the estimator is constructed properly using augmented
139 inverse probability weighting or equivalent approaches.

140 ¹⁵ proved that for a large class of commonly used regression models – including linear and
141 logistic regression – standard hypothesis tests using robust (sandwich) variance estimators
142 are guaranteed to have correct Type I error asymptotically, even when the regression model
143 is misspecified. This provides theoretical justification for the common practice of fitting a
144 potentially misspecified working model and using robust standard errors.

¹⁶ extended these ideas to binary outcomes using targeted maximum likelihood estimation (TMLE), showing that the marginal risk difference can be estimated efficiently by fitting a logistic regression model and then marginalizing (standardizing) over the covariate distribution. This approach avoids the well-known problem identified by¹⁷ that covariate adjustment in logistic regression can reduce precision of the conditional odds ratio. As¹⁶ and¹³ later clarified, this problem does not arise when the target is the marginal effect.

2.2.4 Survival Outcomes

Covariate adjustment in the survival analysis setting has its own theoretical development.¹⁸ showed how auxiliary covariates can be incorporated into the log-rank test to improve efficiency.¹⁹ proposed a nonparametric covariate-adjusted log-rank test with a simple closed form that achieves guaranteed efficiency gains over the unadjusted test and applies universally across simple and covariate-adaptive randomization schemes.²⁰ developed improved estimators for the restricted mean survival time that leverage baseline covariates without assuming proportional hazards.

2.3 Regulatory Guidance

2.3.1 ICH E9 (1999) and E9(R1) (2019)

The ICH E9 guideline²¹ advises investigators to ‘identify covariates expected to have an important influence on the primary variable’ and ‘specify how to account for these in the analysis, in order to improve the precision of the treatment comparisons.’ The E9(R1) addendum²² introduced the estimand framework, which requires precise definition of the treatment effect of interest. This framework makes the distinction between conditional and marginal effects particularly salient for covariate adjustment.

2.3.2 EMA Guideline (2015)

The European Medicines Agency guideline²³ provides concrete recommendations: (i) the primary analysis should include only pre-specified covariates, (ii) no treatment-by-covariate interaction terms should appear in the primary model, (iii) stratification variables used in the randomization should generally be included as covariates, and (iv) covariates should be measured before randomization.

2.3.3 FDA Guidance (2023)

The FDA’s final guidance on covariate adjustment²⁴ represents the most comprehensive regulatory position to date. Key recommendations include:

- Pre-specification of covariates in the statistical analysis plan
- Use of ANCOVA for continuous outcomes
- For nonlinear models (logistic, Cox), use of standardization (g-computation) to estimate marginal effects
- Use of robust (sandwich) or bootstrap standard errors
- Inclusion of stratification factors as covariates

- Avoidance of data-driven covariate selection

The FDA guidance was informed directly by the semiparametric theory of¹³ and the practical work of^{25, 26}, and others.

2.4 The Conditional versus Marginal Distinction

A critical conceptual issue for nonlinear models is the distinction between conditional and marginal treatment effects. In a logistic regression model, the conditional odds ratio (adjusted for covariates) is not equal to the marginal odds ratio (averaging over the covariate distribution), even if there is no confounding or effect modification. This ‘noncollapsibility’ has generated confusion about whether covariate adjustment ‘changes the estimand.’

²⁷ addressed this directly, arguing that if the target parameter is carefully defined, adjustments can be used with minimal assumptions and the benefit of increased efficiency is achieved at essentially no cost.²⁸ reviewed the available methods and advocated for the standardization approach, which leverages the covariate-outcome relationship for efficiency while targeting a well-defined marginal effect.²⁹, in a commentary on this work, agreed that covariate adjustment ‘accounts for outcome heterogeneity and prevents power loss,’ but cautioned against the marginal/standardization framing, arguing that conditional models preserve the full information content of the covariates and that marginal effects can obscure treatment effect heterogeneity.

2.5 Seven Myths and Persistent Misconceptions

³⁰ enumerated seven common misconceptions about randomization, several of which bear directly on covariate adjustment:

1. *Balance is necessary for valid inference.* False: randomization justifies inference regardless of balance; adjustment is for efficiency, not validity.
2. *Observed covariates may be ignored because one has randomized.* False: ignoring prognostic covariates wastes information and reduces power.
3. *Randomization is inefficient.* False: combined with appropriate analysis, randomization plus covariate adjustment is highly efficient.

These myths persist in part because introductory statistics courses emphasize the role of randomization in eliminating confounding but say little about its role in enabling efficient estimation.

2.6 Empirical Evidence on Efficiency Gains

Multiple empirical studies have quantified the practical benefits of covariate adjustment.³¹ showed via simulation that pre-specified adjustment for prognostic covariates in trials with binary outcomes can reduce required sample sizes by 3% to 46%, depending on the strength of the covariate-outcome association.³² demonstrated a 25% reduction in sample size requirements in traumatic brain injury (TBI) trials when adjusting for seven baseline predictors in the IMPACT dataset.³³ reanalyzed the CRASH trial ($n = 9497$), finding that covariate

adjustment using validated prognostic models substantially increased power.³⁴ showed that the difference between adjusted and unadjusted effects can be substantial in the presence of outcome heterogeneity, with adjustment providing a more accurate estimate of the treatment benefit.

²⁶ demonstrated efficiency gains equivalent to a 22-30% reduction in sample size in simulations.²⁵ applied semiparametric methods to COVID-19 treatment trials, achieving meaningful precision gains for binary, ordinal, and time-to-event outcomes.³⁵ proposed a method for adjusting via a prognostic score trained on external (historical) data, showing that the approach attains the semiparametric efficiency bound under certain conditions.

2.7 Practical Guidance for Covariate Selection

⁷ provided the first systematic framework for selecting covariates. Their key recommendations are: (i) covariates balanced by design (stratification factors) should always be included, (ii) strongly prognostic covariates should be included regardless of balance, (iii) data-dependent selection methods based on the current trial data should be avoided as they produce misleading inference, and (iv) all decisions should be pre-specified in the protocol based on prior knowledge.

³⁶ offered a comprehensive practical guide for planning covariate adjustment in individually randomized trials. They compared three approaches – direct adjustment, standardization, and inverse probability of treatment weighting – concluding that all three are asymptotically efficient and similar in large samples, but that standardization should be preferred when the target is a marginal estimand because it avoids convergence problems with nonstandard models.

³⁷ addressed the question of how many covariates to include, showing that the benefits of adjustment diminish and can reverse when too many weakly prognostic covariates are added relative to the sample size.

³⁸ examined adjustment in nonlinear models, confirming that the standard advice for linear models does not always transfer and that care is needed with the choice of estimand and variance estimation.

³⁹ provided a synthesis linking covariate-adaptive randomization, stratification, and covariate-adjusted analysis, with practical recommendations for modern trial design and analysis.

2.8 Present Study

The preceding literature provides a strong theoretical and empirical foundation, and several sources already give practitioner-facing recommendations that overlap substantially with what follows, including the FDA and EMA guidance documents themselves,³⁶'s practical guide to planning adjustment,⁴⁰'s empirical assessment of the consequences of (not) adjusting, and²⁸'s review of conditional-versus-marginal considerations. This paper does not claim a new empirical finding or a novel estimator; its contribution is a single, citation-backed decision procedure that maps each of these dispersed sources onto three operational questions – when to adjust,

which covariates to adjust for, and how to specify the adjustment – together with a worked illustration of why the common test-then-adjust heuristic fails. We synthesize the evidence into these decision rules and illustrate the efficiency and coverage consequences through a small Monte Carlo simulation.

3 Methods

3.1 Simulation Design

We conduct two Monte Carlo simulations, both reported following the ADEMP framework of¹ (Aims, Data-generating mechanisms, Estimands, Methods, Performance measures). The first illustrates the efficiency gains from ANCOVA relative to the unadjusted comparison of means for a continuous outcome, across a range of covariate-outcome correlations, and shows that testing for baseline balance and adjusting only when a significant imbalance is detected (the ‘test-then-adjust’ strategy) does not improve – and can distort the coverage of – inferential properties. It varies the covariate-outcome correlation (γ) and sample size (n) only; it does not vary baseline imbalance as a design factor, and it is restricted to a single continuous outcome and a correctly specified adjustment model (see Limitations). The second scenario (Binary Outcome Under Model Misspecification, below) adds a binary outcome and a deliberately misspecified working model, to illustrate criteria 8 and 10.

3.2 Data Generating Model

For each replication $b = 1, \dots, B$:

1. Generate n subjects, randomized 1:1 to treatment ($Z = 1$) or control ($Z = 0$).
2. Generate a baseline covariate $X_i \sim \text{Normal}(0, 1)$.
3. Generate the outcome as

$$Y_i = \beta_0 + \tau Z_i + \gamma X_i + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

where $\tau = 0.3$ is the true treatment effect, γ controls the strength of the prognostic covariate, and $\sigma^2 = 1 - \gamma^2$ so that $\text{Var}(Y) = 1$ marginally.

3.3 Analysis Strategies

For each simulated dataset, we apply three analysis strategies:

1. **Unadjusted:** Two-sample t -test comparing treatment group means.
2. **Always-adjust (ANCOVA):** Linear regression of Y on Z and X , with robust (HC2) standard errors.
3. **Test-then-adjust:** First test $H_0 : \bar{X}_1 = \bar{X}_0$ at $\alpha = 0.10$. If rejected, use ANCOVA; otherwise use the unadjusted analysis.

Table 1: Simulation parameters.

Parameter	Values
Sample size (n)	100, 200
Treatment effect (τ)	0.3
Covariate effect (γ)	0, 0.3, 0.5, 0.7
Residual variance (σ^2)	$1 - \gamma^2$
Number of replications (B)	5000

3.4 Parameter Values

3.5 Metrics

For each scenario and analysis strategy, we compute:

- **Bias:** Mean $\hat{\tau} - \tau$
- **Empirical SE:** Standard deviation of $\hat{\tau}$ across replications
- **Mean model SE:** Mean of estimated standard errors across replications
- **Coverage:** Proportion of 95% CIs containing τ
- **Power:** Proportion of replications rejecting $H_0 : \tau = 0$ at $\alpha = 0.05$ (two-sided)

3.6 Binary Outcome Under Model Misspecification

The scenario above covers only a continuous outcome under a correctly specified linear model, leaving criterion 8 (standardization for binary outcomes) and criterion 10 (robustness to misspecification) without simulation support. We add a second scenario to address this directly.

3.6.1 Data Generating Model

For each replication, n subjects are randomized 1:1 to $Z \in \{0, 1\}$ and a baseline covariate $X_i \sim \text{Normal}(0, 1)$ is generated. The binary outcome is

$$\text{logit}\{P(Y_i = 1)\} = \beta_0 + \tau Z_i + \gamma X_i + \gamma_2 X_i^2,$$

with $\beta_0 = -0.5$, $\tau = 0.6$ (log-odds treatment effect), and $\gamma = 0.8$ (linear log-odds covariate effect). The quadratic term $\gamma_2 X_i^2$ is the true data-generating nonlinearity; it also defines the misspecification, since one of the two working models below omits it entirely. The estimand is the marginal risk difference $E[Y(1)] - E[Y(0)]$, obtained once by numerical integration of the true model over the known X density (not estimated from simulated data), because τ is a conditional log-odds parameter that does not by itself equal the marginal risk difference under a nonlinear model.

3.6.2 Analysis Strategies

1. **Unadjusted:** difference in sample proportions between arms, with the usual normal-approximation standard error.
2. **Standardized (correct):** fit $\text{logit}\{P(Y_i = 1)\} = b_0 + b_1 Z_i + b_2 X_i + b_3 X_i^2$ by maximum likelihood, predict $P(Y_i = 1)$ under the counterfactuals $Z_i = 1$ and $Z_i = 0$ for every

subject, average each set of predictions over the observed covariate distribution, and difference the two averages (marginal standardization / g-computation). The standard error is obtained by the delta method, combining a numerical gradient of the standardized contrast with an HC2 sandwich covariance matrix for (b_0, b_1, b_2, b_3) .

3. **Standardized (misspecified)**: identical procedure, but the working model omits the quadratic term ($\text{logit}\{P(Y_i = 1)\} = b_0 + b_1 Z_i + b_2 X_i$), so the fitted conditional model is wrong whenever $\gamma_2 \neq 0$.

Both standardization estimators target the same marginal risk difference; per¹⁵ and¹⁶, this target should remain identifiable from a possibly misspecified working model in a randomized trial, because randomization ensures the covariate distribution is the same, in expectation, across counterfactual arms, so averaging model predictions over the observed (not a covariate-adjusted) distribution removes the dependence on correct specification of the conditional mean.

Table 2: Binary-outcome simulation parameters.

Parameter	Values
Sample size (n)	100, 200
Intercept (β_0)	-0.5
Treatment effect, log-odds (τ)	0.6
Linear covariate effect, log-odds (γ)	0.8
Quadratic covariate effect / misspecification (γ_2)	0, 0.4, 0.8
Number of replications (B)	2000

4 Results

4.1 Efficiency Gains from ANCOVA

Table 3 presents the simulation results (all figures computed directly from the saved simulation output; none are hard-coded). We observe that when the covariate is nonprognostic ($\gamma = 0$), all three strategies perform similarly, as expected. As γ increases, ANCOVA achieves progressively smaller empirical standard errors and correspondingly higher power. For $\gamma = 0.7$ (i.e., the covariate explains 49% of outcome variance), ANCOVA increases power from 0.32 to 0.55 at $n = 100$, and from 0.57 to 0.84 at $n = 200$.

The test-then-adjust strategy's power (0.32 at $n = 100$, $\gamma = 0.7$) is close to the unadjusted power and far below ANCOVA's, so it is more accurate to say test-then-adjust behaves like the unadjusted analysis on power rather than as an intermediate strategy. Its coverage, however, is systematically above nominal: at $n = 100$, $\gamma = 0.7$, coverage is 0.969 (Monte Carlo SE 0.002), and at $n = 200$ it is 0.971, both several Monte Carlo standard errors above 0.95. This is not model selection bias in the usual sense; it reflects that most replications are approximately balanced, so test-then-adjust falls back to the unadjusted standard error (mean model SE 0.200 at $n = 100$, $\gamma = 0.7$), which is conservative relative to the empirical variability that would obtain if ANCOVA were always used (0.177 empirical SE for test-then-adjust versus 0.195 mean model SE in the same scenario). The strategy is therefore strictly dominated by

Table 3: Simulation results across analysis strategies and covariate strengths ($B = 5000$ replications per scenario). MCSE columns are Monte Carlo standard errors for coverage and power, following the ADEMP framework of Morris, White, and Crowther (2019).

n	γ	Strategy	Bias	Emp. SE	Mean SE	Coverage	MCSE(Cov.)	Power	MCSE(Pow.)
100	0.0	Unadjusted	-0.002	0.197	0.199	0.952	0.003	0.310	0.007
100	0.0	ANCOVA	-0.002	0.199	0.200	0.952	0.003	0.307	0.007
100	0.0	Test-then-adjust	-0.002	0.198	0.200	0.951	0.003	0.310	0.007
200	0.0	Unadjusted	0.000	0.139	0.141	0.955	0.003	0.562	0.007
200	0.0	ANCOVA	0.000	0.140	0.142	0.956	0.003	0.557	0.007
200	0.0	Test-then-adjust	0.000	0.140	0.141	0.956	0.003	0.560	0.007
100	0.3	Unadjusted	0.004	0.199	0.199	0.951	0.003	0.328	0.007
100	0.3	ANCOVA	0.003	0.191	0.191	0.951	0.003	0.351	0.007
100	0.3	Test-then-adjust	0.003	0.196	0.199	0.953	0.003	0.324	0.007
200	0.3	Unadjusted	0.000	0.139	0.141	0.954	0.003	0.568	0.007
200	0.3	ANCOVA	0.000	0.134	0.135	0.951	0.003	0.600	0.007
200	0.3	Test-then-adjust	0.000	0.136	0.141	0.956	0.003	0.570	0.007
100	0.5	Unadjusted	0.001	0.201	0.200	0.953	0.003	0.322	0.007
100	0.5	ANCOVA	0.001	0.176	0.174	0.947	0.003	0.404	0.007
100	0.5	Test-then-adjust	0.003	0.190	0.198	0.961	0.003	0.327	0.007
200	0.5	Unadjusted	0.002	0.143	0.141	0.947	0.003	0.566	0.007
200	0.5	ANCOVA	0.002	0.124	0.123	0.945	0.003	0.681	0.007
200	0.5	Test-then-adjust	0.002	0.136	0.139	0.957	0.003	0.579	0.007
100	0.7	Unadjusted	0.000	0.202	0.200	0.946	0.003	0.316	0.007
100	0.7	ANCOVA	0.000	0.145	0.143	0.947	0.003	0.551	0.007
100	0.7	Test-then-adjust	-0.001	0.177	0.195	0.969	0.002	0.322	0.007
200	0.7	Unadjusted	0.003	0.142	0.141	0.949	0.003	0.569	0.007
200	0.7	ANCOVA	0.000	0.102	0.101	0.948	0.003	0.837	0.005
200	0.7	Test-then-adjust	0.002	0.125	0.137	0.971	0.002	0.600	0.007

always-adjust: it sacrifices power relative to ANCOVA without offering a validity advantage, since ANCOVA's coverage is close to nominal in every scenario studied.

4.2 Relative Efficiency

We see in Figure 1 that the relative efficiency of ANCOVA increases approximately as $1/(1 - \gamma^2)$, consistent with theory. This ratio is asymptotically independent of sample size; the close agreement between the $n = 100$ and $n = 200$ curves here is consistent with that asymptotic result but is based on only two sample sizes and should not be read as a general empirical finding across arbitrary n .

4.3 Coverage Properties

Figure 2 shows that the unadjusted and ANCOVA strategies maintain approximately nominal coverage throughout. The test-then-adjust strategy shows systematic over-coverage that grows with γ , as quantified in Table 3 and discussed above; this is a consequence of the SE mixture described above, not a form of model-selection bias in the estimate itself (the point estimates remain approximately unbiased in every scenario, per Table 3).

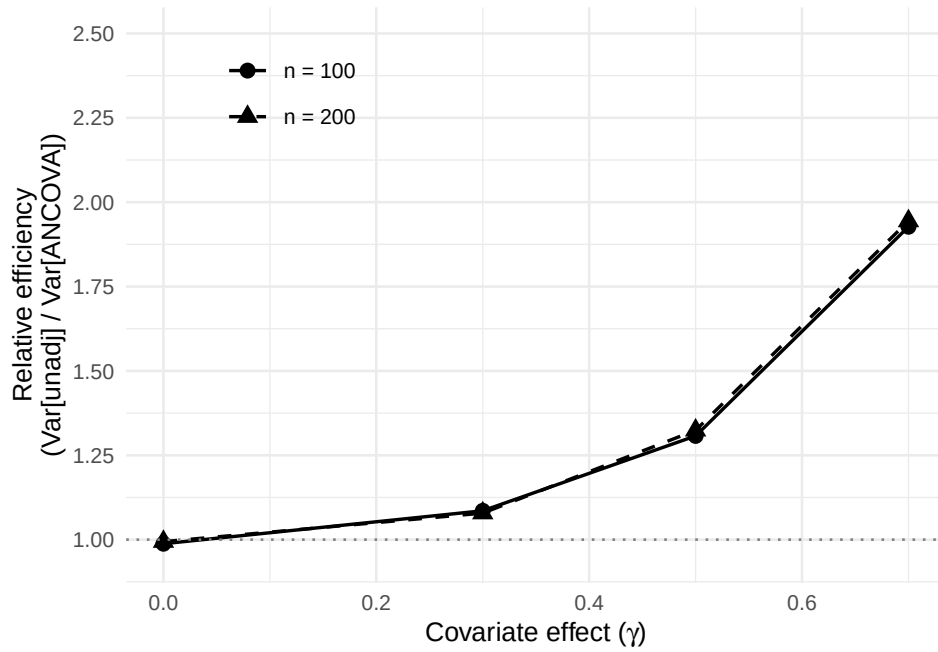


Figure 1: Relative efficiency of ANCOVA versus the unadjusted estimator as a function of covariate prognostic strength. Relative efficiency is defined as the ratio of empirical variances (unadjusted / ANCOVA).

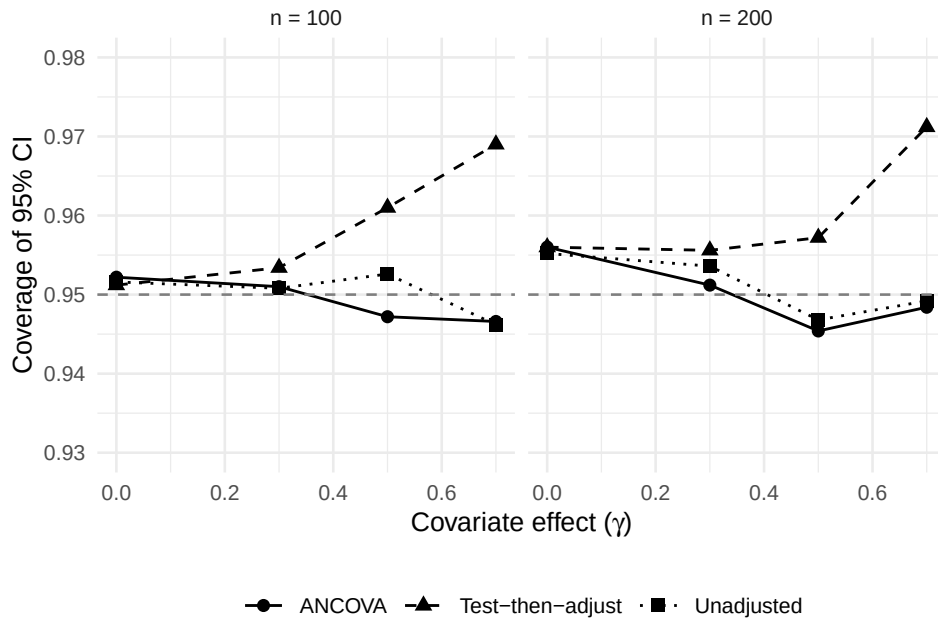


Figure 2: Empirical coverage of 95% confidence intervals across analysis strategies. The horizontal dashed line marks nominal 95% coverage.

Table 4: Binary-outcome simulation results across analysis strategies and misspecification magnitudes ($B = 2000$ replications per scenario). Bias, coverage, and power are relative to the true marginal risk difference, obtained by numerical integration of the true data-generating model. MCSE columns are Monte Carlo standard errors, following the ADEMP framework of Morris, White, and Crowther (2019).

n	γ_2	Strategy	Bias	Emp. SE	Mean SE	Coverage	MCSE(Cov.)	Power	MCSE(Pow.)
100	0.0	Unadjusted	-0.003	0.098	0.098	0.941	0.005	0.258	0.010
100	0.0	Standardized (correct)	-0.002	0.092	0.093	0.943	0.005	0.276	0.010
100	0.0	Standardized (misspecified)	-0.002	0.092	0.093	0.943	0.005	0.283	0.010
200	0.0	Unadjusted	0.001	0.069	0.070	0.953	0.005	0.470	0.011
200	0.0	Standardized (correct)	0.001	0.065	0.066	0.948	0.005	0.512	0.011
200	0.0	Standardized (misspecified)	0.001	0.065	0.066	0.948	0.005	0.510	0.011
100	0.4	Unadjusted	0.003	0.100	0.098	0.942	0.005	0.278	0.010
100	0.4	Standardized (correct)	0.002	0.095	0.094	0.949	0.005	0.301	0.010
100	0.4	Standardized (misspecified)	0.003	0.096	0.095	0.948	0.005	0.294	0.010
200	0.4	Unadjusted	0.000	0.070	0.070	0.948	0.005	0.478	0.011
200	0.4	Standardized (correct)	0.000	0.066	0.066	0.949	0.005	0.504	0.011
200	0.4	Standardized (misspecified)	0.000	0.067	0.067	0.948	0.005	0.495	0.011
100	0.8	Unadjusted	0.000	0.098	0.097	0.947	0.005	0.260	0.010
100	0.8	Standardized (correct)	0.000	0.092	0.091	0.945	0.005	0.288	0.010
100	0.8	Standardized (misspecified)	0.001	0.096	0.096	0.946	0.005	0.256	0.010
200	0.8	Unadjusted	0.002	0.071	0.069	0.945	0.005	0.461	0.011
200	0.8	Standardized (correct)	0.003	0.066	0.064	0.939	0.005	0.510	0.011
200	0.8	Standardized (misspecified)	0.003	0.069	0.068	0.941	0.005	0.481	0.011

4.4 Binary Outcome and Model Misspecification

Table 4 presents the binary-outcome results. At $n = 200$ and the largest misspecification magnitude studied ($\gamma_2 = 0.8$), the misspecified standardization estimator's bias is 0.0027, close to the correctly specified estimator's 0.0028 and small relative to the true marginal risk difference; its coverage is 0.941 (Monte Carlo SE 0.005), consistent with nominal 95%. This supports the theoretical claim, discussed under Theoretical Foundations and reiterated in criterion 10, that standardization from a possibly misspecified working model remains valid for the marginal treatment effect in a randomized trial: omitting the true quadratic covariate term does not reintroduce meaningful bias here. The cost is in efficiency rather than validity: the empirical SE of the misspecified standardization estimator (0.0692) falls between the correctly specified estimator's (0.0664) and the unadjusted estimator's (0.0706), so misspecification narrows but does not eliminate the efficiency advantage of standardization over the unadjusted difference in proportions. When $\gamma_2 = 0$ the two standardization working models coincide and their estimates are numerically close, as expected.

5 Discussion

5.1 Summary of Objective Criteria

Based on the theoretical literature, empirical evidence, and regulatory guidance, we propose the following objective criteria for covariate adjustment in the primary analysis of an RCT.

When to adjust:

1. *Always*, if prognostic covariates are available and pre-specified. The decision to adjust should never be based on observed baseline balance^{2,30}.
2. *Always* include stratification variables used in the randomization design as covariates in the analysis^{6,23,24}.
3. *Always* include the baseline measurement of the outcome variable, if available⁸⁻¹⁰.

Which covariates to adjust for:

4. Select covariates based on prior knowledge of their prognostic value – from previous trials, meta-analyses, or established clinical understanding^{7,37}.
5. The number of covariates should be small relative to the sample size.⁷ provide a formula relating the benefit of adjustment to the number of subjects per covariate.
6. Do not select covariates based on the trial data itself (neither on baseline imbalance nor on observed covariate-outcome associations). Data-driven selection produces biased inference^{4,7}.

How to adjust:

7. For continuous outcomes: use ANCOVA (linear regression of Y on Z and X). Standard (OLS) or robust standard errors are both acceptable for balanced designs^{12,24}.
8. For binary outcomes: fit a logistic regression model and compute the marginal effect (standardization / g-computation) to obtain a risk difference or relative risk^{16,28}, a procedure illustrated directly in the Binary Outcome and Model Misspecification results above. Alternatively, use a linear probability model or modified Poisson regression with robust standard errors.
9. For time-to-event outcomes: use a covariate-adjusted log-rank test^{18,19} or estimate the restricted mean survival time with covariate adjustment²⁰.
10. Use robust (sandwich) standard errors to guard against model misspecification^{13,15,24}. The Binary Outcome and Model Misspecification results above illustrate this directly: a standardized estimator built from a working model that omits a true nonlinear covariate term remains approximately unbiased for the marginal risk difference, at some efficiency cost.
11. Pre-specify all covariates and the analysis method in the statistical analysis plan. The primary analysis should contain no interaction terms²³.

5.2 Consistency with Regulatory Guidance

These criteria are fully consistent with the EMA guideline²³ and the FDA final guidance²⁴. Both agencies endorse pre-specified covariate adjustment as a tool for improving precision and recommend against data-driven covariate selection. The FDA guidance specifically recommends standardization for nonlinear models and robust variance estimation.

5.3 Limitations

The simulations presented here are deliberately simple and illustrative, not a comprehensive evaluation of the eleven criteria. Both are restricted to a single covariate and no missing data, and neither varies baseline imbalance as a design factor. The continuous-outcome scenario

uses a correctly specified linear data-generating model throughout. The binary-outcome scenario adds one misspecified working model (a quadratic term omitted from the covariate) and one comparator estimator (standardization), which supports criteria 8 and 10 but does not exhaust them: it does not examine alternative link functions, multiple covariates, more severe or qualitatively different (e.g., non-monotone, interaction-driven) forms of misspecification, or the propensity-score and augmented estimators discussed in the Theoretical Foundations section. Neither scenario includes a time-to-event outcome, so criterion 9 (survival outcomes) still rests on the cited theoretical and applied literature rather than on evidence generated here. Real trials involve multiple covariates, various outcome types, missing data, and complications such as noncompliance and intercurrent events. The estimand framework of ICH E9(R1)²² introduces additional considerations about what exactly is being estimated. The core qualitative message – that pre-specified covariate adjustment improves efficiency without sacrificing validity, and that standardization remains valid under working-model misspecification in a randomized trial – is supported by the broader literature cited throughout this paper across outcome types and regulatory frameworks. The two scenarios simulated here demonstrate it directly for a correctly specified continuous-outcome model and for a binary outcome under one form of working-model misspecification; the time-to-event case (criterion 9) remains supported only by the cited literature, not by simulation evidence generated here.

The simulation uses heteroskedasticity-consistent (HC2) standard errors for the ANCOVA and test-then-adjust strategies, computed via `sandwich::vcovHC()`, matching the Methods description and the FDA’s recommendation for robust variance estimation. For the balanced, correctly-specified designs considered here HC2 and classical OLS standard errors are asymptotically equivalent¹²; the sandwich estimator was retained because it remains valid under the misspecification and heteroscedasticity that can arise in practice, even though neither is present by construction in this simulation.

6 Future Research

Several directions for future investigation emerge from this review:

1. **Machine learning for covariate adjustment.** The FDA guidance does not yet address the use of machine learning methods for constructing prognostic scores.³⁵ showed that a prognostic score trained on external data can achieve the efficiency bound, but the operating characteristics of cross-validated or penalized approaches in the trial data itself remain under study.⁴¹ explored combining covariate adjustment with group sequential and information-adaptive designs.
2. **Missing covariate data.** When baseline covariates are missing, the analyst faces a tradeoff between the efficiency gains from adjustment and the potential bias from ad hoc handling of missingness. Multiple imputation and augmented estimators are active research areas.
3. **Cluster-randomized and crossover trials.** The theory reviewed here applies primarily to individually randomized, parallel-group trials. Extensions to cluster random-

ization and crossover designs present additional methodological challenges.

4. **Software tools.** The R package `estimatr` implements Lin’s regression estimator via the `lm_lin()` function. Broader software implementing the standardization approach for various outcome types would make best practices more accessible.

5. **Harmonization of conditional and marginal frameworks.** The tension between conditional effects (natural for clinicians reasoning about individual patients) and marginal effects (natural for regulators reasoning about populations) remains unresolved^{27–29}.

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The simulation function, the driver script that runs it, the saved simulation output, and the manuscript source are all in the `baselinedist` project repository. Specific paths are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x. The simulation function `run_sim()` lives in `R/run_sim.R` as a documented, exported package function and is unit-tested in `inst/tinytest/test_run_sim.R`. It uses base R `lm()` for model fitting, `sandwich::vcovHC()` (HC2) for the ANCOVA and test-then-adjust standard errors, `tibble` for the returned summary, `dplyr` and `ggplot2` for reporting, and `rmarkdown` plus `bookdown` for the manuscript build. `MASS` is not used; the single covariate is generated with `stats::rnorm()`. The binary-outcome scenario’s simulation function `run_sim_binary()` lives alongside it in `R/run_sim_binary.R`, is unit-tested in `inst/tinytest/test_run_sim_binary.R`, and uses base R `glm(family = binomial())` for model fitting, `sandwich::vcovHC()` (HC2) combined with a numerical (central finite-difference) gradient for the delta-method standard error of the standardized contrast, and `stats::integrate()` to compute the true marginal risk difference from the known data-generating model.

Random-number generator. `analysis/scripts/run_simulation.R` calls `run_sim(..., seed = master_seed + scenario_id)` once per scenario, with `master_seed <- 20260305`, so each of the eight (n, γ) scenarios has its own reproducible seed set at the top of `run_sim()`. There is no per-replicate reseeding within a scenario; replicates within a scenario share one continuous RNG stream started from that scenario’s seed.

analysis/scripts/run_simulation_binary.R follows the same pattern with its own master_seed <- 20260819, so the six (n, γ_2) binary-outcome scenarios each have their own reproducible seed, independent of the continuous-outcome scenario's RNG stream.

Data and code availability. Neither simulation is embedded in this Rmd. analysis/scripts/run_simulation runs the continuous-outcome factorial design ($n \in \{100, 200\}$, $\gamma \in \{0, 0.3, 0.5, 0.7\}$, $B = 5000$) and saves the combined results, together with the master seed and a timestamp, to analysis/data/derived_data/sim_results.rds. analysis/scripts/run_simulation_binary.R runs the binary-outcome factorial design ($n \in \{100, 200\}$, $\gamma_2 \in \{0, 0.4, 0.8\}$, $B = 2000$) and saves its results to analysis/data/derived_data/sim_results_binary.rds. This document's simulation and simulation-binary chunks read those .rds files directly and stop with an informative error if either is missing, so the manuscript can never silently render with stale or absent results. To regenerate the results, run Rscript analysis/scripts/run_simulation.R (approximately one to two minutes on a laptop) and Rscript analysis/scripts/run_simulation_binary.R (approximately four to six minutes, since glm() fitting and the delta-method standard error are substantially more expensive per replicate than the continuous scenario's lm() fits) from the repository root; then re-render this document with bash tools/render.sh analysis/report/report.Rmd. The manuscript source, bibliography (references.bib), and CSL file are at analysis/report/.

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